INDIVIDUAL WEEKLY PROJECT REPORTS

Student: Annalakshmi R

Role: Data Preprocessing and Exploratory Data Analysis (EDA)

Project Title: Flight Delay Analysis and Visualization

WEEK 5 – Exploratory Data Analysis: Understanding Flight Delays

During the fifth week, I focused on exploratory data analysis to understand the patterns present in the cleaned flight dataset. EDA helped me examine the characteristics of the data and identify useful information before moving towards advanced statistical techniques.

I concentrated on important delay-related variables, particularly departure delay and arrival delay. I examined how these variables could be used to understand the frequency and extent of flight delays.

I also explored the possibility of grouping flight information based on airlines and airports. Such comparisons can help identify

differences in delay patterns across different categories. I understood that grouping and summarizing the data could provide useful insights that would not be easily visible from individual records.

Another important aspect of my work was understanding the distribution of numerical variables. I examined how measures such as mean, median, minimum, and maximum values could help describe the delay data.

I coordinated with Dharshini to discuss the visual representations required for presenting the EDA findings. This helped connect the analytical work with the visualization stage of the project.

Outcome and Learning:

By the end of this week, I had gained practical knowledge of exploratory data analysis and understood how descriptive statistics and grouping techniques could help reveal patterns in flight delay data.